DTC INSPECTION [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

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Note
- If the self-test is implemented, the seat weight sensor calibration is performed. Perform the DTC inspection in accordance with the following conditions:
  — Do not place objects on the front seat (passenger side)
  — Do not touch the front seat (passenger side)
  — Fix the slide position of the front seat (passenger side) to the maximum rearward position.
  — Fix the recliner position of the front seat (passenger side) to the maximum forward position.
  — Set cabin vehicle temperature range between the **-10 to 40 °C {14 to 104 °F}**
  — Other than the above conditions, a “Japan only” command is also displayed, however, do not select the command because it is for use in Japan only.

CMDTC Self Test
1. Connect the M-MDS to the DLC-2.
2. After the vehicle is identified, select the following items from the initialization screen of the M-MDS.
   (1) Select "Self Test".
   (2) Select "All CMDTCs".
3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection. (See DTC TABLE [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)
4. After completion of repairs, clear all DTCs stored in the SAS control module. (See CLEARING DTC [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)

ODDTC Self Test
1. Connect the M-MDS to the DLC-2.
2. After the vehicle is identified, select the following items from the initialization screen of the M-MDS.
   (1) Select "Self Test".
   (2) Select "Modules".
   (3) Select "RCM".
3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection. (See DTC TABLE [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)
4. After completion of repairs, clear all DTCs stored in the SAS control module. (See CLEARING DTC [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)

Snapshot Data Table

Note
- The SAS control module stores the following two types of information when a DTC is detected and displays snap shot data in the M-MDS.
  — Vehicle information detected by SAS control module
  — Vehicle information detected by instrument cluster and received by SAS control module via CAN signal

- The data for all DTCs currently detected is stored.

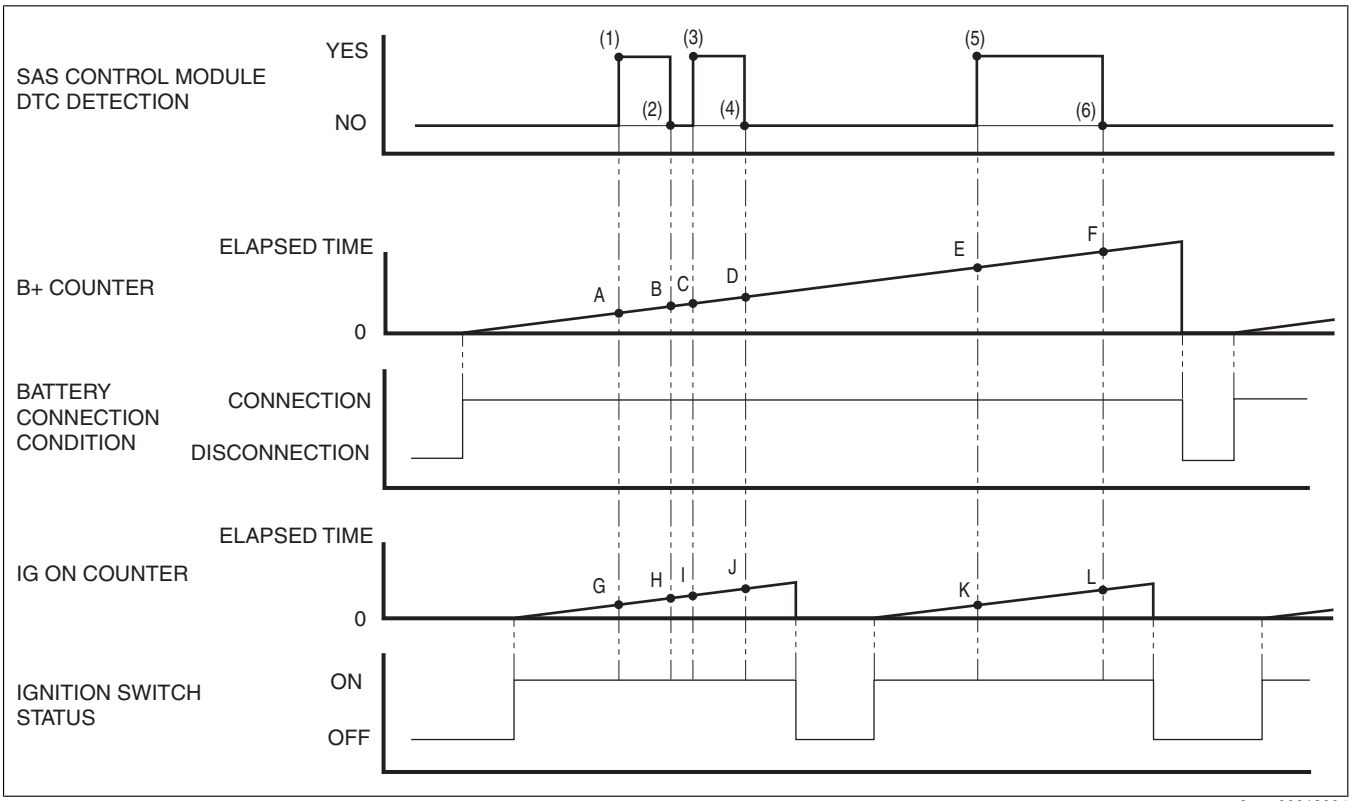
Snapshot data item	Unit		Definition	Data read/use method	Corresponding PID data monitor item
AAT	°C	°F	Ambient temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under20%/ Over20%/ FAIL		Accelerator pedal position status	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error		Instrument cluster configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0-Under 80 degrees C/ Over 80 degrees C/ FAIL		Engine coolant temperature status	—	—
FAULT_CNT	—		Number of malfunction detections	—	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding PID data monitor item
FIRST_DET_IG	hh:mm:ss*1	Time when initial malfunction occurs (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects first malfunction*3	—	—
IC_VPWR	V	Instrument cluster power supply voltage	• The SAS control module constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	VPWR*2
IG-ON_TIMER	hh:mm:ss*1	Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The SAS control module constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
LAST_CLR	hh:mm:ss*1	Repair time for last malfunction (B+ counter) • Elapsed time from when battery is connected until SAS control module detects that last malfunction is repaired*3	—	—
LAST_CLR_IG	hh:mm:ss*1	Repair time for last malfunction (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects that last malfunction is repaired*3	—	—
LAST_DET	hh:mm:ss*1	Time when last malfunction occurs (B+ counter) • Elapsed time from when battery is connected until SAS control module detects last malfunction*3	—	—

Snapshot data item	Unit		Definition	Data read/use method	Corresponding PID data monitor item
LAST_DET_IG	hh:mm:ss*1		Time when last malfunction occurs (IG-ON counter) • Elapsed time from when ignition is switched ON (engine off or on) until SAS control module detects last malfunction*3	—	—
PWR_MODE_KEY	Key Out/ Key Recently Out (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 2)/ Ignition On (Position 2)/ Running (Position 2)/ Running - Starting		• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time **within 3 s** since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time **within 3 s** since ignition was switched ON (engine off or on) • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The SAS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the ignition switch status when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
RPM_STATUS	Engine Stop/ Under1500rpm/ Over1500rpm/ FAIL		Engine speed status	• The SAS control module constantly receives the engine speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the engine speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	TACHOMTR*2
SHIFT_STATUS	P/N/ D/ R/ FAIL		Selector lever position status	• The SAS control module constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the selector lever position when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
TOTAL_DIST	km	miles	Accumulated total traveled distance from completion of vehicle until SAS control module detects DTC (Odometer value in instrument cluster)	The total traveled distance from which the SAS control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding PID data monitor item
TOTAL_TIME	hh:mm:ss*1	Accumulated total elapsed time since vehicle completion until SAS control module detects a DTC*3 **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time from which the SAS control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the instrument cluster PID item TOTAL_TIME. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*2
VPWR	V	SAS control module power supply malfunction	—	VPWR_IGA
VSPD_STATUS	Stop/ 0-10km/h/ Over10km/h/ FAIL	Vehicle speed status	• The SAS control module constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the SAS control module records the vehicle speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	SPEEDOMTR*2

*1 : The seconds may be indicated after the decimal point.
*2 : Instrument cluster PID (See PID/DATA MONITOR TABLE [INSTRUMENT CLUSTER].)
*3 : The snap shot data of the number of stored malfunction detections, B+ counters, or IG ON counters is combined with the DTC detection conditions of the SAS control module, and the point-in-time data in the following table is stored.



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SAS control module DTC detection period	Snapshot data						
	FAULT_CNT (Number of malfunction detections)	**TOTAL_TIME**	**FIRST_DET_IG**	**LAST_DET**	**LAST_DET_IG**	**LAST_CLR**	**LAST_CLR_IG**
	Number of malfunction detections	**Time when first malfunction occurs (B+ counter)**	**Time when initial malfunction occurs (IG-ON counter)**	**Time when last malfunction occurs (B+ counter)**	**Time when last malfunction occurs (IG-ON counter)**	**Repair time for last malfunction (B+ counter)**	**Repair time for last malfunction (IG-ON counter)**
(1)	1	A	G	A	G	-	-
(2)	1	A	G	A	G	B	H
(3)	2	A	G	C	I	B	H
(4)	2	A	G	C	I	D	J
(5)	3	A	G	E	K	D	J
(6)	3	A	G	E	K	F	L